

Module 1: Probability and Random Variables

Syllabus

MODULE 1 Probability and Random Variables	Introduction to conditional probability, independent events, Total probability, Baye's theorem and problems, Navie Bayes concept, Definition and examples of random variables, Discrete & continuous random variables, Expectation of a random variable, mean and variance of a random variable.
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1.1 CONDITIONAL PROBABILITY

Problems:

1. If A and B are two events with $P(A) = 1/2$, $P(B) = 1/3$ and $P(A \cap B) = 1/4$, compute the following
(i) $P(A/B)$ ii) $P(B/A)$ iii) $P(\bar{A}/\bar{B})$ iv) $P(\bar{B}/\bar{A})$
2. In a college 25% students are failed in Mathematics, 15% in Chemistry and 10% students failed in both Mathematics and Chemistry. A student is selected at random.
(i) If he failed in Chemistry what will be the probability that he will fail in Mathematics
(ii) If he failed in Mathematics what will be the probability that he will fail in Chemistry
(ii) What will be the probability that he fails in either of the two subjects.
3. The probability that three students A, B, C solve a problem are $\frac{1}{2}$, $\frac{1}{3}$, $\frac{1}{4}$ respectively. Iff problem is simultaneously assigned to all of them, what is the probability that the problem is solved?
4. The probability that a team wins a match is $\frac{3}{5}$. If this team play 3 matches in a tournament, what is the probability that the team
i) Win all the matches ii) win atleast one match
iii) win atmost one match iv) win all the matches

Answers: 1) $\frac{3}{4}$ $\frac{1}{2}$ $\frac{5}{8}$ $\frac{5}{6}$ 2) $\frac{2}{3}$ $\frac{2}{5}$ $\frac{3}{10}$ 3) $\frac{3}{4}$ 4) $\frac{27}{125}$ $\frac{17}{125}$ $\frac{44}{125}$ $\frac{8}{125}$

Practice Problems:

1. If A and B are events with $P(A) = \frac{3}{8}$, $P(B) = \frac{5}{8}$ and $P(A \cup B) = \frac{3}{4}$, find $P(A/B)$ and $P(B/A)$ Ans: i) $\frac{2}{5}$ ii) $\frac{2}{3}$
2. Consider randomly selecting a student at a certain university, and let A denote the event that the selected individual has a Visa credit card and B be the analogous event for a Master Card. Suppose that $P(A) = 0.5$, $P(B) = 0.5$ and $P(A \cap B) = 0.25$. Calculate and interpret each of the following probabilities
i) $P(B/A)$ ii) $P(B'/A)$ iii) $P(A'/B)$
Given that the selected individual has at least one card, what is the probability that he or she has a Visa card? Ans: i) 0.5 ii) 0.5 iii) 0.5 iv) 0.67
3. A certain shop repairs both audio and video components. Let A denote the event that the next component brought in for repair is an audio component, and let B be the event that the

next component is a compact disc player (so the event B is contained in A). Suppose that $P(A) = 0.6$ and $P(B) = 0.05$. Then find $P(B/A)$. Ans: 0.0833

4. In a group of 300 all adults, 272 are right-handed, 3 adults are selected with replacement. Find a) $P(\text{all 3 are right-handed})$ b) $P(\text{all 3 left-handed})$ c) $P(\text{At least one right-handed})$

Ans: i) 0.745 ii) 0.0008 iii) 0.9992

5. Three students A,B,C write an examination. Their chances of passing are $\frac{1}{4}$, $\frac{1}{2}$ and $\frac{1}{3}$ respectively. Find the probability that (i) all of them pass, (ii) at least one of them passes, (iii) at least two of them pass. Ans: $\frac{3}{4}$

6. A solar water heater company manufactures two parts namely the heating panel and the insulated tank. In the manufacturing process 9% are likely to be defective in the panel and 5% are likely to be defective in the insulated tank. If an assembled unit is installed in a house what is the probability that it is non defective. Ans: 0.8645

1.2 Total Probability

Problems:

1. A company producing electric relays has three manufacturing plants producing 50, 30, and 20 percent, respectively, of its product. Suppose that the probabilities that a relay manufactured by these plants is defective are 0.02, 0.05, and 0.01, respectively. If a relay is selected at random from the output of the company, what is the probability that it is defective? Ans: 0.027

2. Two numbers are chosen at random from among the numbers 1 to 10 without replacement. Find the probability that the second number chosen is 5. Ans: 0.1

3. A person has undertaken a mining job. The probabilities of completion of the job on time with and without rain are 0.42 and 0.90 respectively. If the probability that it will rain is 0.45, then determine the probability that the mining job will be completed on time.

Ans: 0.684

Practice Problems

1. If 40% of boys opted for maths and 60% of girls opted for maths, then what is the probability that maths is chosen if half of the class's population is girls? Ans: 0.5
2. Company A produces 10% defective products, Company B produces 20% defective products and C produces 5% defective products. If choosing a company is an equally likely event, then find the probability that the product chosen is defective. Ans: 0.12
3. Suppose 5 men out of 100 men and 10 women out of 250 women are colour blind, then find the total probability of colour blind people. (Assume that both men and women are in equal numbers.) Ans: 0.045
4. The probability that the political party A does a particular work is 30% and the political party B doing the same work is 40%. Then find the probability that the work is completed if the probability of choosing the political party A is 40% and that of B is 60%.

Ans: 0.36

1.3 Baye's Theorem

Problems:

1. Three Machines A, B, C produces 50%, 30%, 20% of the total production of a factory. The percentage of defective outputs are 3, 4, 5. If a product is selected at random, then what is the probability that it is defective? What is the probability that it is from A?
Ans: i) 0.037 ii) 0.4054
2. In a bolt factory there are four machines A, B, C, D manufacturing respectively 20%, 15%, 25%, 40% of the total production. Out of these 5%, 4%, 3%, 2% are defective. If a bolt is drawn at random was found defective, then what is the probability that it was manufactured by A or D?
Ans: i) 0.0315 ii) 0.5715
3. Three major parties A, B, C are contending for power in the elections of a state and the chance of their winning the election is in the ratio 1:3:5. The parties A, B, C respectively have probabilities of banning the online lottery $\frac{2}{3}$, $\frac{1}{3}$, $\frac{3}{5}$. What is the probability that there will be a ban on the online lottery in the state? What is the probability that the ban is from the party C?
Ans: 0.6429
4. Ball pen refills are packed in pouches containing 25 refills in each pouch. In a shop it was found that 5 refills failed to write in pouch 1, 10 each in pouches 2 and 3, 1 refill in pouch 4. Suppose a refill is selected at random from one of the four pouches, what is the probability that it fails to write? What is the probability that it was from pouch 4?
Ans: i) 0.26 ii) 0.04

Practice Problems:

1. The chance that a doctor will diagnose a disease correctly is 60%. The Chance that a patient will die after correct dragonize is 40% and the chance of death by wrong diagnosis is 70%. If a patient dies, what is the chance that his disease was correctly diagnosed?
Ans: 0.4615
2. In a class 70% are boys and 30% are girls .5% of boys, 3% of the girls are irregular to the classes. What is the probability of a student selected at random is irregular to the classes and what is the probability that the irregular student is a girl?
Ans:i)0.044 ii)0.2045
3. An Office has 4 secretaries handling respectively 20%,60%,15% and 5% of the files of all government reports. The probability that they misfile such reports are respectively 0.05 ,0.1,0.1 and 0.05. Find the probability that the misfiled report can be blamed on the first secretary.
Ans: 0.1143
4. In a college where boys and girls are in equal proportion, it was found that 10 out of 100 boys and 25 out of 100 girls were using the same brand of a two wheeler. If a student using that was selected at random what is the probability of being a boy?
Ans: i) 0.175 ii) 0.2857
5. A certain firm has plants A, B and C producing, respectively, 35%, 15% and 50%, of the total out. The probabilities of a non-defective product are, respectively, 0.75,0.95 and0.85. If a consumer receives a defective product, what is the probability that it came from plant C.
Ans: i) 0.17 ii) 0.4412

6. the chances that a politician, a business man or an academician will be appointed as a vice-chancellor (V.C) of a university are 0.5, 0.3, 0.2 respectively. Probability that research is promoted by these persons if they are appointed as V.C are 0.3, 0.7, and 0.8 respectively.
- (i) Determine the probability that research is promoted.
 - (ii) If the research is promoted, what is the probability that V.C is an academician?

Ans: i) 0.66 ii) 0.6061